

Data Flow Diagram

Date	2 July 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	2 Marks

Data Flow Diagram

The diagram below is the Level-1 Data Flow Diagram (DFD) for the HDI Prediction System (Human Development Intelligence). It shows how data moves through the system — between the user, the internal processes, and the data stores — as a country's indicators are turned into a predicted Human Development Index classification (Very High, High, Medium, or Low).

DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded shape	External Entity	A person, organization, or system outside the project that sends or receives data (e.g., User, Policymaker, Researcher).
Rectangle with numbered header	Process	An activity that transforms incoming data into outgoing data (e.g., Preprocess & Normalize Data, Predict HDI Score).
Rectangle (solid fill, no number)	Data Store	A place where data is held for later use (e.g., Country Indicator Dataset, Trained HDI Prediction Model).
Labeled arrow	Data Flow	The movement of data between entities, processes, and data stores, labeled with what data is being passed.

HDI Prediction System — Level-1 DFD

A user (member of the public, policymaker, or researcher) selects a country and supplies its development indicators — life expectancy, mean and expected years of schooling, and GNI per capita (PPP). Process 1.0 validates this input. Process 2.0 preprocesses and normalizes it against the reference indicator dataset (D1). Process 3.0 runs the normalized feature vector through the trained prediction model (D2) to compute an HDI score. Process 4.0 classifies the score into a development tier, logs the result to the prediction history (D3), and returns the score, tier, and insights to the user.

Level-1 Data Flow Diagram — HDI Prediction System (Human Development Intelligence)

